

PINKSYNC: BIDIRECTIONAL SIGN LANGUAGE AUTHENTICATION AND COMMUNICATION SYSTEM

PATENT CLAIMS

1. A bidirectional communication and authentication system comprising:
 - An optical sensor array for capturing three-dimensional sign language movements and non-manual markers;
 - A facial feature tracking module configured to identify and track at least 43 facial points for grammatical and emotional expression recognition;
 - A spatial grammar processing engine that maps visual-spatial grammar to linear language structures and vice versa;
 - An authentication module that verifies user identity through unique sign language biometrics; and
 - A bidirectional translation system that converts between sign language and spoken/written language.
2. The system of claim 1, wherein the authentication module comprises:
 - A sign biometric verification system that authenticates users based on individualized signing patterns;
 - A motion-based password alternative system that replaces text passwords with sign sequences;
 - A visual two-factor authentication system that combines traditional authentication with sign verification; and
 - A cross-platform authentication persistence system that maintains authenticated status across multiple devices.
3. The system of claim 1, further comprising a cultural integration layer including:
 - A regional sign variant recognition module capable of identifying and processing dialectal variations in sign languages;

- A community-specific terminology database that maintains specialized vocabulary used within Deaf communities;
- A visual vernacular translation system that preserves the richness of visual storytelling; and
- A Deaf epistemology modeling system that incorporates Deaf ways of knowing and processing information.

4. The system of claim 1, wherein the spatial grammar processing engine comprises:

- A three-dimensional coordinate mapping system that tracks signing space utilization;
- A visual cognitive load management system that optimizes information presentation based on visual processing capacity;
- A non-manual signal recognition system that identifies and interprets facial expressions as grammatical markers; and
- A classifier prediction system that interprets descriptive classifiers in sign language.

5. The system of claim 1, further comprising a multi-modal visual communication platform that:

- Supports multiple sign languages;
- Facilitates visual-spatial information organization;
- Enables persistent visual communication across devices; and
- Incorporates Deaf cultural norms in interface design.

6. A method for authenticating users through sign language biometrics, comprising:

- Capturing a sequence of sign language movements using optical sensors;
- Analyzing the movements against stored biometric profiles;
- Verifying user identity based on signing style, rhythm, and spatial parameters;
- Generating an authentication token upon successful verification; and
- Maintaining authentication persistence across an ecosystem of connected devices.

7. A method for bidirectional translation between sign languages and spoken languages, comprising:

- Capturing sign language input through optical sensors;
- Processing non-manual markers including at least 43 facial feature points;
- Analyzing spatial grammar patterns in three-dimensional space;
- Converting visual-spatial grammatical structures to linear language structures;
and
- Generating synthetic voice output that preserves semantic and emotional content.

8. The method of claim 7, further comprising:

- Capturing spoken language through audio sensors;
- Analyzing linguistic structures of spoken input;
- Converting linear language structures to visual-spatial grammatical structures;
and
- Generating visual representation guidance for sign language output.

9. A system for visual-spatial information architecture, comprising:

- A spatial organization interface that presents information in three-dimensional visual space;
- A visual memory mapping system that organizes information according to Deaf cognitive patterns;
- A visual search engine optimized for spatial recall rather than text-based queries;
and
- A collaborative visual workspace that enables multiple users to interact in shared visual space.

10. The system of claim 9, wherein the visual-spatial information architecture further includes:

- An eye tracking system that monitors visual attention patterns;
- A cognitive load optimizer that adjusts information density based on visual processing capacity;
- A visual notification system designed according to Deaf attention management principles; and
- A visual history tracking system that enables spatial recall of previous

interactions.